SIEMENS

Sensation

CT

Troubleshooting Guide

WCS (water/air)

Troubleshooting Guide WCS (water/air)

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Safety information

Read the following instructions as well as the most recent version of

- **General Safety Notes** (TD00-000.860.01.xx.xx)
- **Product- specific safety Information** (CT02-023.860.01.xx.xx)

before trouble shooting the WCS.

The following instructions and the above mentioned **Safety documentation** contain important safety information as well as information about tools to be used.

NOTE

Only use the <Read> button in the functions <Table load/modify>. All other functions are not necessary for service, and are normally for development staff only.

=> Improper handling can result in system malfunctions.

NOTE

Operating the WCS water pump without water (e.g. during filling/refilling the gantry) will destroy the pump.

=> Do not operate the water pump without water!

NOTE

Do not operate the system in customer mode at the increased water pressure as used for the test. Always return the water pressure to the nominal operating pressure of 2.0 bar ±0.2 bar, with the heat exchanger switched off.

Tools

- Standard tool kit
- 2 water hoses with 1/2"" connections for filling and draining the system
- Manometer (included with the delivery of WCS)
- Absorbent cloth
- Blotting paper

Overview

The Water Cooling System has to ensure the proper temperature inside the gantry. Therefore, errors caused by a defective WCS may generate:

- temperature error messages for components within the gantry
- temperature error messages for the MCU
- image quality problems

This section provides additional troubleshooting information, in case information in either the detailed message or functional descriptions is insufficient.

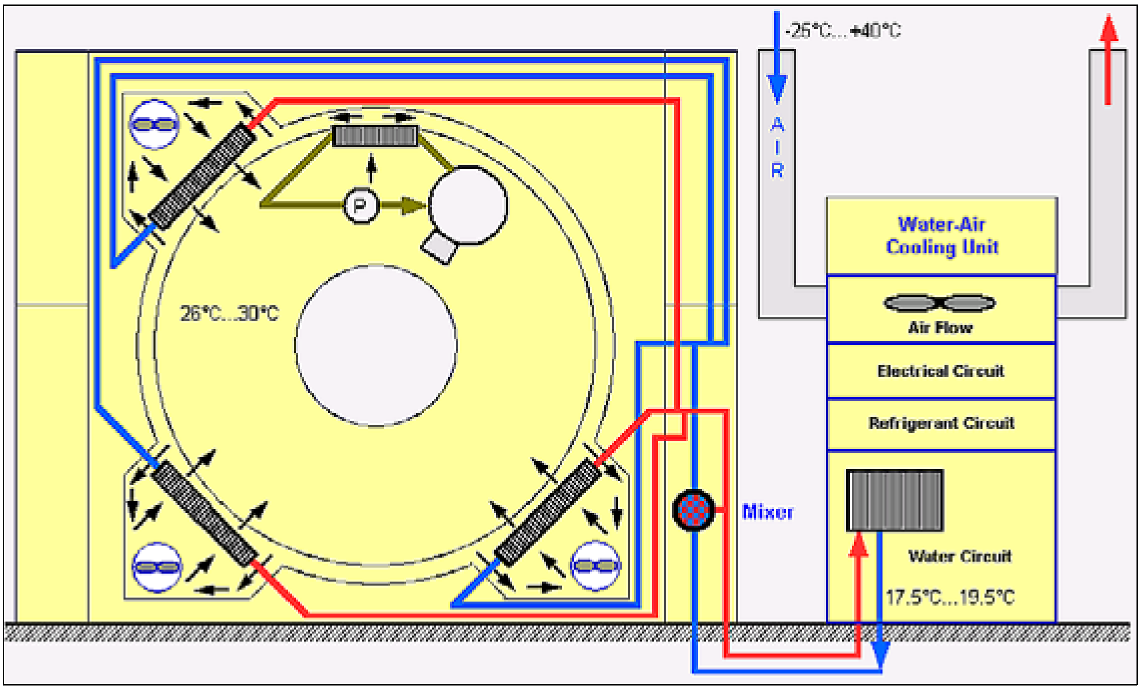

Fig. 1: Cooling circuit overview

Temperature value

Nominal temperature value

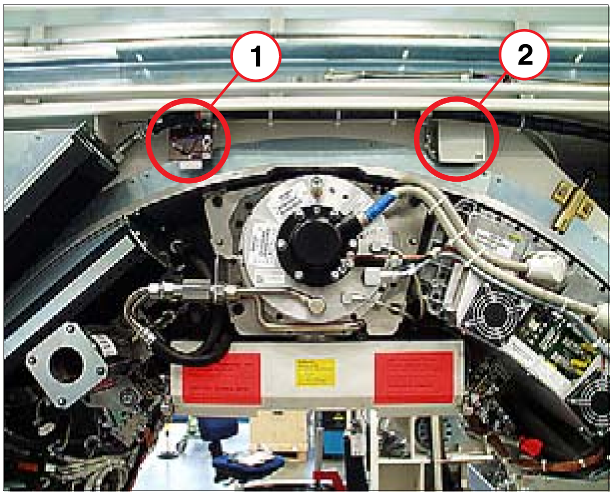

Fig. 2: Temperature Sensor in Gantry

Pos. 1 B 305

Pos. 2 B 302

Internal gantry temperature

- B302 = the actual temperature value, used for regulation.
  B305 = the safety temperature switch. In case of overheating (> 43°C), the PDC switches off the power to PSS.
- Nominal temperature:
  28°C ± 2°C
- Thresholds
  < 35°C (B302)
  < 43°C (B305)

Temperature inside the WCS

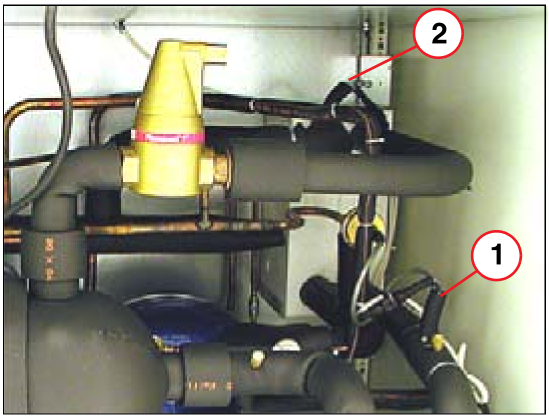

Fig. 3: Temperature Sensor in WCS

Pos. 1	R1
Pos. 2	R3

Fig. 4: Temperature sensor in the air inlet

Pos. 1	clip
Pos. 2	R2

➪ Temperature Sensor in WCS:
R1 = the outlet water temperature value, used for regulation.
R2 = the inlet air temperature value.
R3 = the refrigerant temperature value, used to protect against freezing.

➪ Nominal temperature:
R1 = 17.5°C to 19.5°C.
R2 = -25.0°C to +40°C.

Current temperature values

The current temperature values of the various sensors can be evaluated via the syngo service software.

How to obtain the current temperature values

1. From the user interface, select **Options -> Local Service**.
2. In the authentication page, enter the **Service Password**and Click **OK.**
3. In the Home menu, click the **Control** button.

4. Click **Table load/modify**:

   ➪ The following page appears.

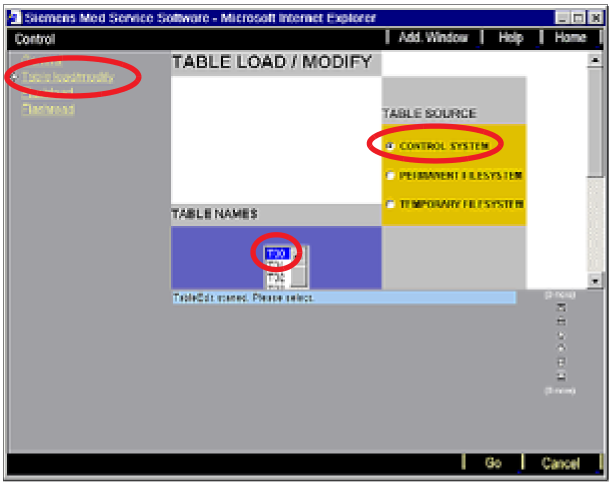

Fig. 5: Table load / modify page

5. To access the control system

   - Select **T00** for TABLE NAME
   - Select **CONTROL SYSTEM** for TABLE SOURCE

   ➪ Click **GO**, and the following page appears.

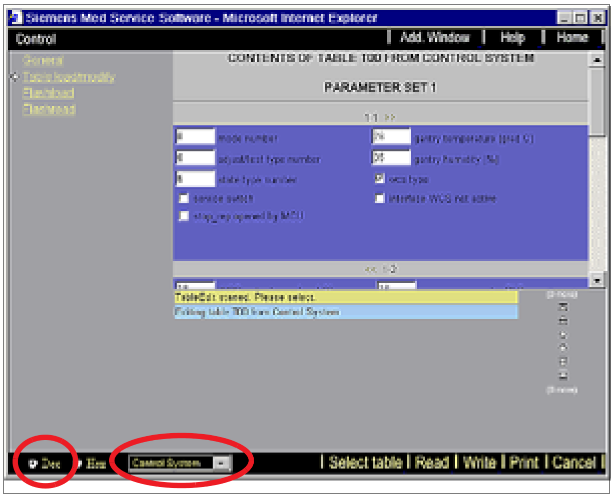

Fig. 6: Example of Parameter Set 1

6. Select **Dec**, and click **Read**:

NOTE	**Only use the <Read> button in the functions <Table load/modify>. All other functions are not necessary for service, and are normally for development staff only.** **=> Improper handling can result in system malfunctions.**

➪ The corresponding table will be activated.
Gantry temp. (B302) = **<gantry temperature>**
Water temperature in secondary cooling circuit (R1) = **<WCS water temp.>**
More important information is available, please refer to (Tab. 2 / p. 15).

If current temperature is out of tolerance

Gantry temperature

Problems may exist with the following:

- Secondary cooling circuit
  - ➪ Refer to (Secondary cooling circuit / p. 10)
- Cooling fan in gantry
  - ➪ Check or replace the fan for the heat exchanger as described in replacement instructions
    CT02-023-841.01.XX.XX
- Temperature sensor (B302)
  - ➪ Check or replace the temperature sensor.

Secondary cooling circuit

Problems may exist with the following:

- WCS
  - ➪ Refer to (Electrical circuit / p. 11)
- Water leaks
  - ➪ Refer(Water circuits / p. 17)
- Air flow
  - ➪ Check the air flow circuits (e.g., radial fan, condenser, inlet/outlet of air, air hoses, air temperature sensor (R2)).
  - ➪ Check or replace the radial fan or the temperature sensor (R2) as described in replacement instructions
    CT02-023-841.06.XX.XX
- Temperature sensor (R1)
  - ➪ Check or replace the temperature sensor (R1) as described in replacement instructions.
    CT02-023-841.06.XX.XX

Electrical circuit

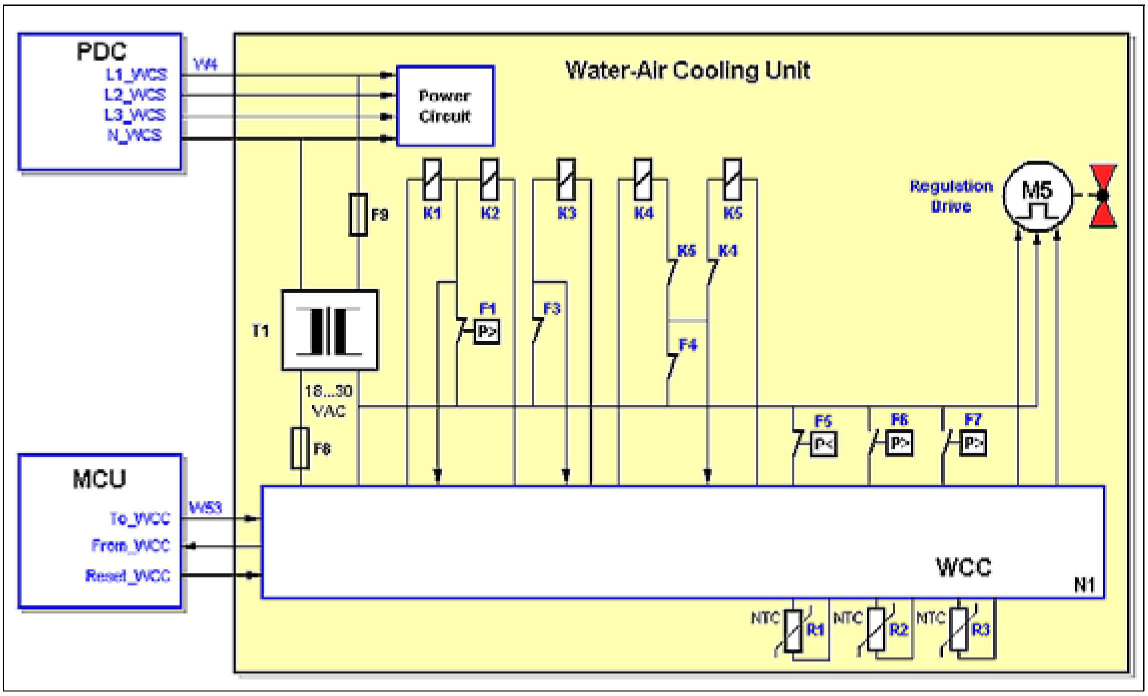

Fig. 7: *WCS Water / Air Electrical Circuit Overview 1*

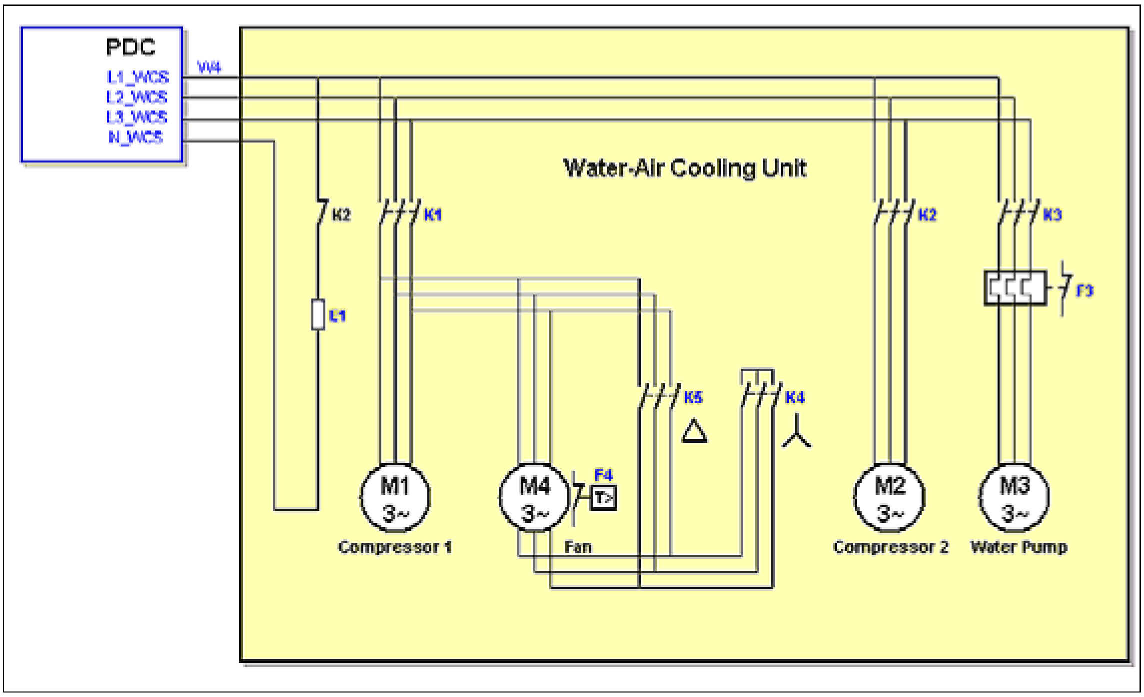

Fig. 8: *WCS Water / Air Electrical Circuit Overview 2*

N1 - WCS controller

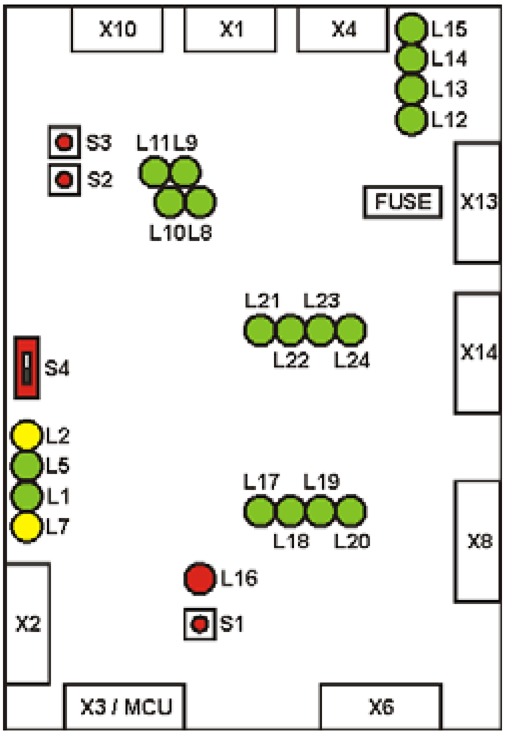

Fig. 9: N1 - WCS Controller (schematic layout)

For further investigations within the WCS, it is necessary to know the current status of the N1 WCS controller. Obtain the actual status from the LED on N1 (LED"s on the N1 controller / p. 12) or the system control table via the service software (The system control table / p. 13).

LED"s on the N1 controller

Tab. 1 LED,s on the N1 controller

LED / color	Normal status	Description for WCS (water-air)
L1 - green	blinking	Program active
L2 - yellow	off	Service active
L3 - n.a.	n.a.	not mounted
L4 - n.a.	n.a.	not mounted
L5 - green	on, if -->	Communication active (telegram traffic)
L6 - n.a	n.a.	not mounted
L7 - yellow	off	on, if --> malfunction
L8 - green	on, if -->	Condensation pressure2 is reached
L9 - green	on, if -->	Condensation pressure1 is reached

LED / color	Normal status	Description for WCS (water-air)
L10 - green	on	off, if pressure too low (pressure in coolant circuit)
L11 - green	on	off, if pressure too high (pressure in coolant circuit)
L12 - green	on, if -->	50Hz off, if --> 60Hz
L13 - green	n.a.	not used
L14 - green	on	Cooling pump protection F3 / OK off, if --> over current
L15 - green	on	Thermocontact: ventilator
L16 - red	off	on, if --> MCU reset occurs
L17 - green	on, if -->	Compressor 1 active
L18 - green	on, if -->	Compressor 2 active
L19 - green	on	Cooling pump running
L20 - green	n.a.	not used
L21 - green	on, if -->	Ventilator is working in Y
L22 - green	on, if -->	Ventilator is working in D
L23 - green	on, if -->	Cooling liquid regulation: colder
L24 - green	on, if -->	Cooling liquid regulation: warmer

The system control table

1. From user interface, select **Options -> Local Service**.
2. In the authentication page, enter the **Service Password** and Click **OK.**
3. In the Home menu, click the **Control** button.

4. Click **Table load/modify**:
   ➪ The following page appears.

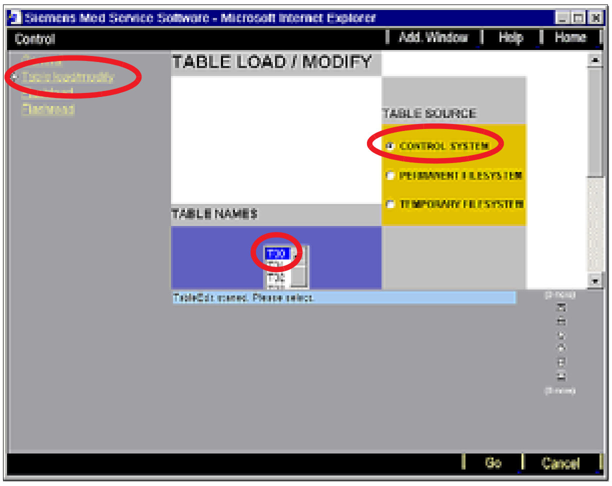

Fig. 10: Table load / modify page

5. To access the control system
   - Select **T00** for TABLE NAME
   - Select **CONTROL SYSTEM** for TABLE SOURCE
   ➪ Click **GO**, and the following page appears

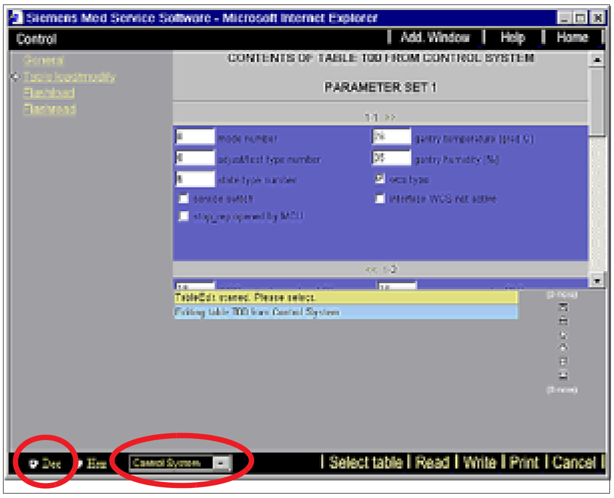

Fig. 11: Example of Parameter Set 1

6. Select **Dec**, and click **Read**
   ➪ The corresponding table will be activated.

NOTE	**Only use the <Read> button in the functions <Table load/modify>. All other functions are not necessary for service, and are normally for development staff only.** **=> Improper handling can result in system malfunctions.**

Tab. 2 T00 table contents

	Normal condition
gantry temperature	current temperature inside gantry
gantry humidity	current humidity inside gantry
WCS water temperature	current temperature in secondary cooling circuit (R1)
compressor 1 active	On, if compressor 1 active
compressor 2 active	On, if compressor 2 active
ventilator Y/D active	On
cooling pump active	On
cooling med out of toler-ance	Off
cooling water out of toler-ance	Off
cooling system out of tol.	Off
cooling system defective	Off
no water flow	Off
motor protector pump	Off
motor protector ventilator	Off
primary cooling media temp	Off / Only for W/W

NOTE

This table T00 include the current temperature value. Please check the tolerance as described in (Temperature value / p. 7).

Other components" parts in electrical circuit

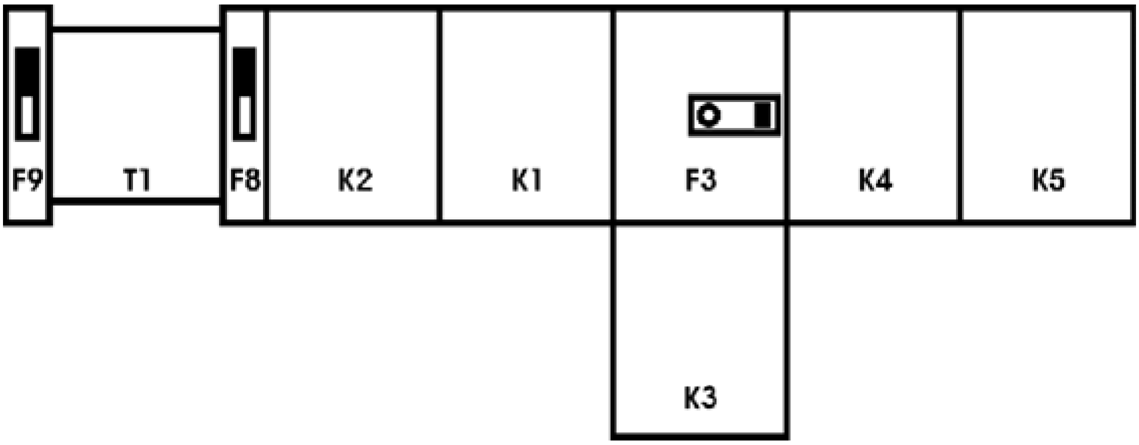

Fig. 12: Other components parts

Tab. 3 Fuse, Measurement points and LED's

	Description
K1	Fan and compressor 1 contactor
K2	Compressor 1 contactor
K3	Cooling pump contactor
K4	Fan star connection contactor
K5	Fan delta connection contactor
F1	Pressure control high protection (refrigerant circuit)
F3	Cooling pump over current protection
F4	Fan motor over current protection
F5	Pressure control low protection (refrigerant circuit)
F6	Fan star connection (slow speed)
F7	Fan delta connection (fast speed)
F8	Circuit breaker 3A
F9	Circuit breaker 1A
T1	Transformer

Water circuits

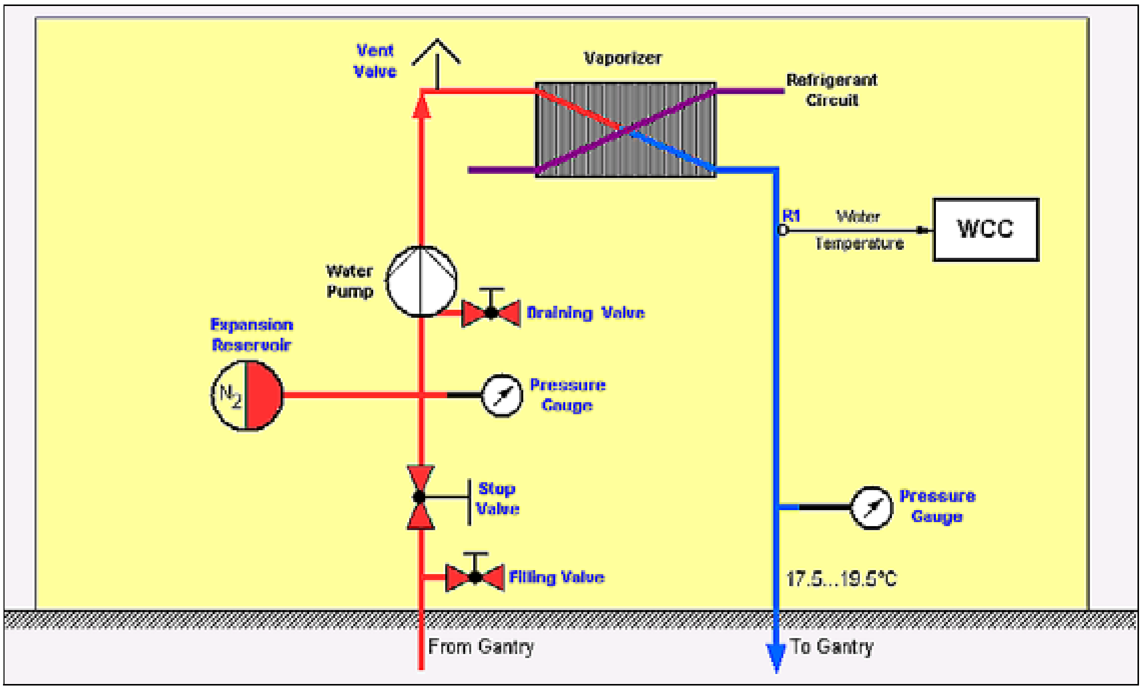

Fig. 13: WCS Water / Air Circuit Overview

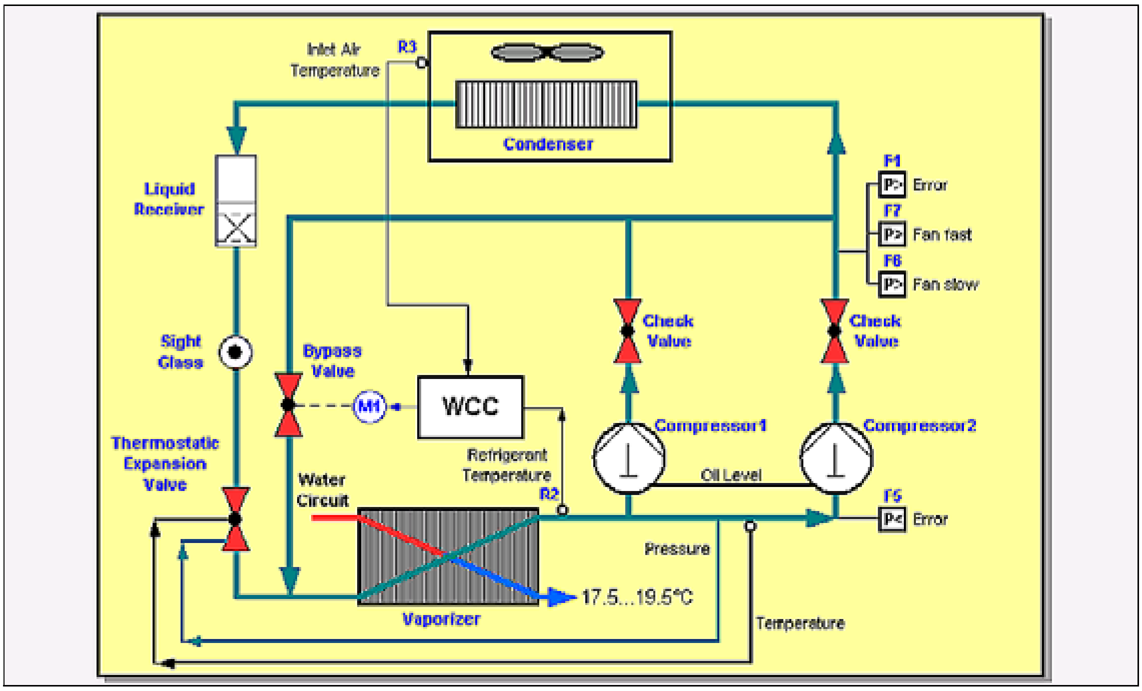

Fig. 14: WCS Water / Air Refrigerant Circuit Overview

Draining and filling the cooling system

Testing for leaks requires increasing or decreasing the water pressure.

Draining process to decrease the water pressure

1. Switch off the WCS via breaker F14 in the PDC.
2. Connect the water hoses to the filling and draining valves (4/Fig. 15 / p. 18).
3. Route the water hoses to a syphon or a bucket.
4. The red shut-off valve for the gantry cooling system must be open.
   - ➪ The lever is lined up with the connection pipe (1/Fig. 15 / p. 18).

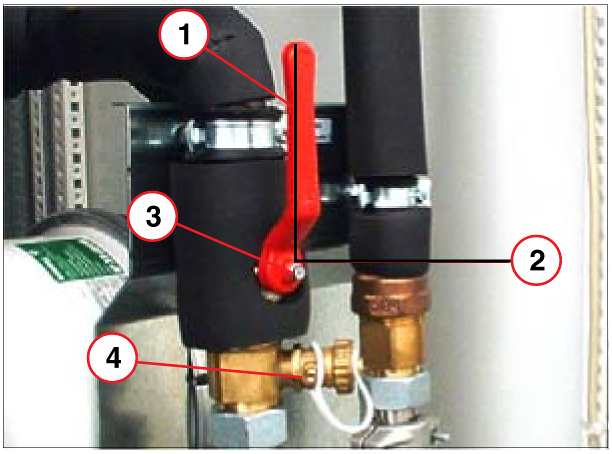

Fig. 15: Shut-off valve at the WCS "Gantry water inlet"

Pos. 1	Open
Pos. 2	Close
Pos. 3	Shut-off valve
Pos. 4	filling and draining valve

5. Connect the manometer at the WCS "gantry water outlet".
6. Open the filling and draining valves (4/Fig. 15 / p. 18) by turning them 90° counterclockwise using the valve cap.
   - ➪ The valve is open when the notch on the valve is lined up with the connection pipe.
7. Drain until the water pressure reaches 2.0 ±0.2 bar, and close the filling and draining valves(4/Fig. 15 / p. 18) by turning them 90° clockwise using the valve cap.
8. Switch off the WCS via breaker F14 in the PDC.

NOTE	**After the WCS has been running for a while, please check that the water pressure is 2.0 ±0.2 bar, with WCS switched off.**

Filling the cooling system

Please always refer to the latest version of

Installation Instructions CT02-023.814.01.xx.xx.: Filling the cooling system.

Reasons for a drop in pressure

After installation

Drop in pressure may occur once or twice after start-up. After filling the system a maximum of 3 times, pressure should remain constant. The pressure should not drop by more than 1.0 bar in 6 months.

Possible causes of pressure drop may include:

- Excessive amounts of air in the cooling system.
- Warm water was used to fill the system (loss in volume when cooled to 18°C).
- The hose for filling the system was not filled with water when connected to the WCS.
- Widening of water hoses within the first weeks after system installation.

After the system has been filled several times

After 6 months the drop in pressure is >1.0 bar, and the minimum value of the static pressure is 1.0 bar.

Possible causes of pressure drop may include:

- Screw connections are not water-tight.
- Leaks in the affected system components (e.g., in the water pump, 3-way valve, Spirovent, expansion reservoir, heat exchanger, connection WCS <--> gantry, within gantry). Continue with (Testing for leaks with WCS switched off / p. 19).

Testing for leaks with WCS switched off

NOTE

Do not operate the system in customer mode at the increased water pressure as used for the test. Always return the water pressure to the nominal operating pressure of 2.0 ± 0.2 bar, with the WCS switched off.

=> Please refer to (Draining process to decrease the water pressure / p. 18).

NOTE

Do not perform scans at this time.

1. Switch off the WCS via breaker F14 in the PDC.
2. Connect the filling hose to the cooling system.
   ➪ Fill the hose with water prior to connecting it to the system to avoid introducing air into the system. Air in the filling hose will lead later to a loss of pressure.
3. Fill it until the pressure reaches a maximum (< 7.5 bar) as described in (Filling the cooling system / p. 18).

NOTE	The maximum pressure depends on the water supplier and the location of the site. In case the pressure is not able to reach 7.5 bar, test will be performed with your maximum.

4. Record the water pressure in the WCS:
   - ➪ If the pressure drops by more than 10% within 1 hour, there is a leak in the system.
5. Inspect the WCS, the gantry, and the full length of all the water hoses for possible leaks (use blotting paper).
6. If leaks are found, replace the corresponding parts. Please refer to the respective replacement instructions:

   Replacement of the Parts: Gantry CT02-023.841.01.xx.xx

   Replacement of the Parts: WCS CT-023.841.06.xx.xx
7. After replacing the parts, make sure that leaks are not found using this test.
8. If leaks are not found with this test, individual components can be tested.

NOTE	**Always return the water pressure to the nominal operating pressure of 2.0 ± 0.2 bar, with the WCS switched off. Please refer to (Draining process to decrease the water pressure / p. 18).**

Testing for leaks in individual WCS components

NOTE	**Do not operate the system in customer mode at the increased water pressure as used for the test. Always return the water pressure to the nominal operating pressure of 2.0 ± 0.2 bar, with the WCS switched off.** **=> Please refer to (Draining process to decrease the water pressure / p. 18).**

NOTE	**Do not perform scans at this time.**

If leaks are not found with the previous test using increased pressure, individual components can be tested as follows.

3-way valve in gantry right stand and WCS

Overview: location of the 3-way valves

Fig. 16: 3-Way Valve in Gantry stand

Proceed as follows:

1. Remove the black insulation covering the 3-way valve.
2. Place absorbent cloth under the 3-way valve, because water may escape during test.
3. Open and close both 3-way valves approx. 10 times. A valve can be opened and closed by turning the white screw of the adjustment drive CW (open) or CCW (closed) with a 5 mm Allen key (1/Fig. 17 / p. 21).

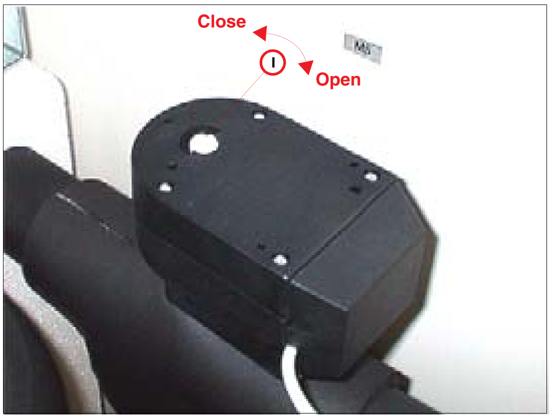

Fig. 17: Example of Adjustment drive

4. Inspect the 3-way valve for leaks (use blotting paper).
5. If leaks are found, replace the corresponding 3-way valve. Please refer to the respective replacement instructions:

   3-Way Valve in Gantry Stand CT02-023.841.01.xx.xx

   3-Way Valve in WCS CT-023.841.06.xx.xx
6. After replacing the 3-way valve, make sure that leaks are not found using this test.
7. Drain until the water pressure reaches 2.0 $\pm$0.2 bar as described in (Draining process to decrease the water pressure / p. 18).
8. Install the black insulation covering the 3-way valve.

NOTE	**Always return the water pressure to the nominal operating pressure of 2.0 ± 0.2 bar, with the WCS switched off. Please refer to (Draining process to decrease the water pressure / p. 18).**

Cooling pump

Overview: location of the devices

00-04/

Fig. 18: Cooling pump

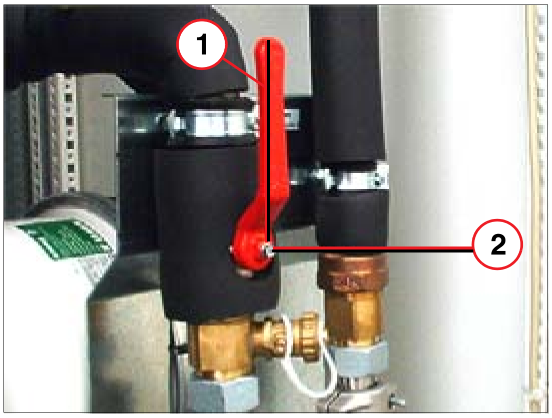

Fig. 19: Shut-off valve

NOTE	**Do not perform the following test with increased water pressure. Even a short-term test with increased water pressure will destroy the water pump. Ensure that the water pressure is the nominal operating pressure of 2.0 ± 0.2 bar, with the WCS switched off.** **=> Please refer to (Draining process to decrease the water pressure / p. 18).**

Proceed as follows:

1. Slide the black insulation sleeves away from the nuts.
2. For approx. 5 minutes, close the shut-off valve (2/Fig. 19 / p. 22) of the cooling pump being tested. This forces the pump to work under increased pressure.
3. Inspect the pump for leaks (use blotting paper).
4. If leaks are found, replace the cooling pump as described in the replacement instructions:

   Replacement of Parts CT-023.841.06.xx.xx
5. After replacing the cooling pump, make sure the leaks are not found with this test.

Additional hints

Pressure Value

- The nominal water pressure should be 2.0 $\pm$0.2 bar, with the WCS switched OFF.
- When the WCS is switched off (pump off), pressure levels at the water inlet (in front of the pump) and at the water outlet (behind the pump) are equal.
- When the WCS is switched on (pump on), a relative pressure difference of approx. 2.5 bar can be measured in front and behind the pump.
- The absolute pressure measured via a manometer depends on the length of the hose and the type of operation (50/60 Hz).

Anti-freeze agent

- It is prohibited to add an anti-freeze agent to the secondary (gantry) cooling circuit.

Others

- Excessive amounts of air in the cooling system (after filling it) cause the pump to overheat, resulting in damage to the motor.
- To operate an additional external water pump or valve, a potential-free contact (max. 1 Amp, 220 Volts) is available in the PDC - PDS.

Chapter	Change	Reason
FMP	SOMATOM 40 was added	136660 (P30G)